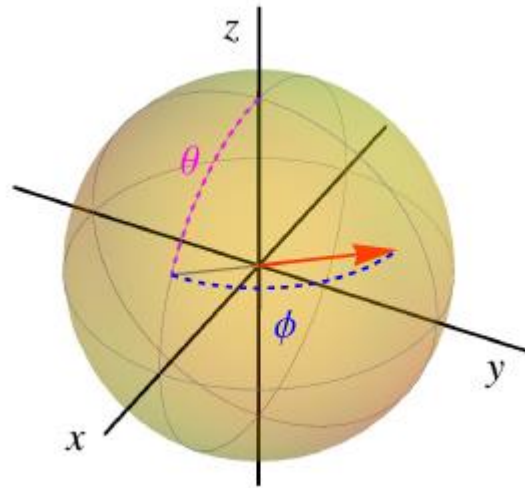


Quantum Computation

ELK336E

Lect #10



A measure for entanglement: Concurrence

Hill S and Wootters W K 1997

Physical Review Letters **78** 5022–5025

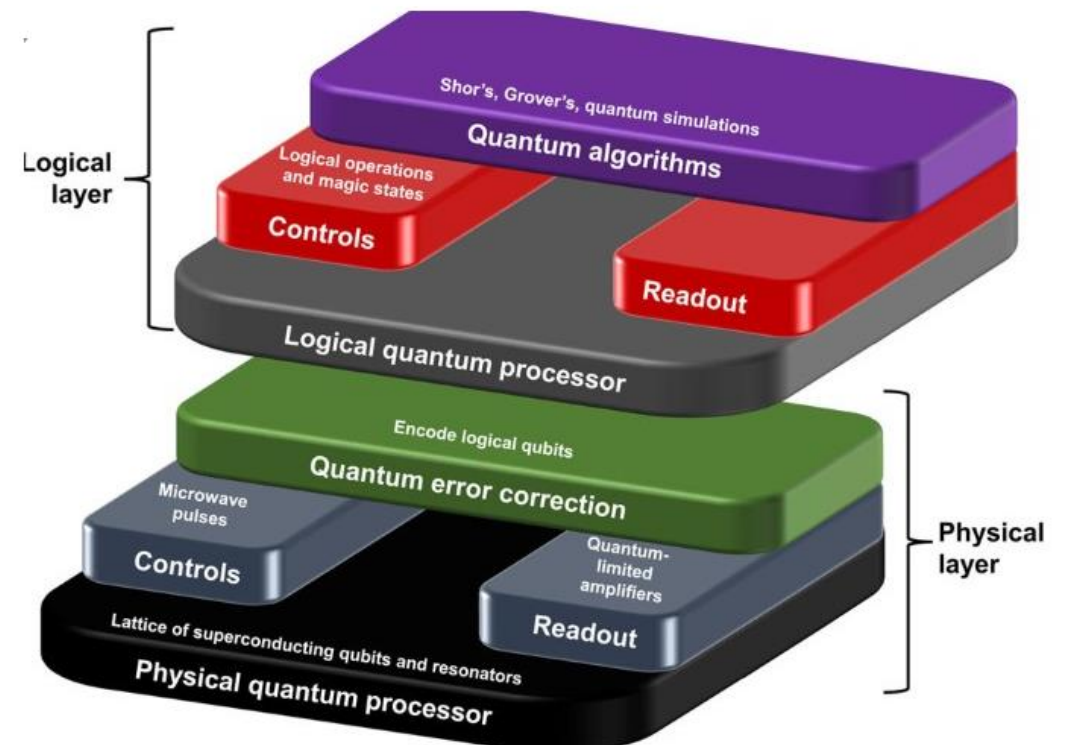
$$C[\rho] = \max\{0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\}$$

Here, $\{\lambda_i\}$ is the square roots of the eigenvalues of $\rho\tilde{\rho}$

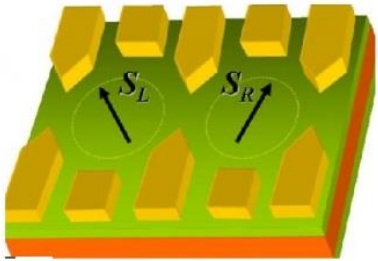
$$\tilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y)$$

and ρ^* is the complex conjugate of ρ .

Quantum hardware

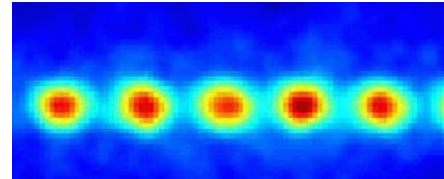


Candidate Systems



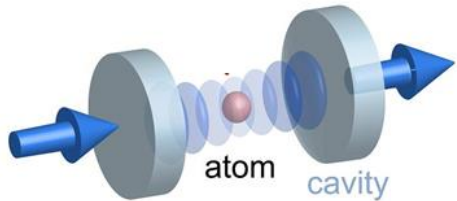
Quantum Dots

Dec time: 10^{-6} sec
Op. Time: 10^{-9} sec



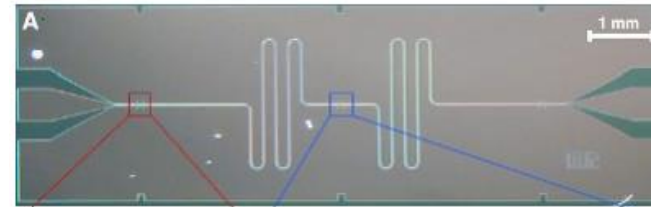
Ion traps

Dec time: 10^{-1} sec
Op. Time: 10^{-14} sec



Cavity QED

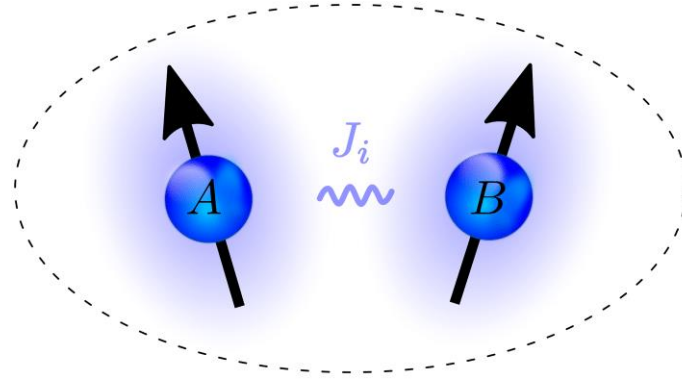
Dec time: 10^{-5} sec
Op. Time: 10^{-14} sec



Superconducting circuits

Dec time: 25×10^{-6} sec
Op. Time: 5×10^{-9} sec

All the physical realizations should mimic the spin interactions.



$$H = H_0 + H_I$$

$$H_0 = \frac{\omega_A}{2} \sigma_z^A + \frac{\omega_B}{2} \sigma_z^B$$

$$H_I = J_x \sigma_x^A \otimes \sigma_x^B + J_y \sigma_y^A \otimes \sigma_y^B + J_z \sigma_z^A \otimes \sigma_z^B$$

If $J_x = J_y = J_z$, this is called a *XXX* type spin interaction model.

If $J_x, J_y = 0$ and $J_z \neq 0$, this is called Ising model.

Now we introduce a new type of operators; Ladder operators;

$$\sigma^+ = \sigma_x + i\sigma_y$$

$$\sigma^- = \sigma_x - i\sigma_y$$

$$\sigma^+ = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\sigma^- = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$|e\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \equiv |0\rangle$$

$$|g\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \equiv |1\rangle$$

$$\begin{array}{c}
 \text{---} |e\rangle \\
 | \hbar\omega \\
 \text{---} |g\rangle
 \end{array}
 \quad
 \begin{array}{ll}
 \sigma^+ |g\rangle = |e\rangle & \sigma^+ |e\rangle = 0 \\
 \sigma^- |e\rangle = |g\rangle & \sigma^- |g\rangle = 0
 \end{array}$$

Note: $\frac{1}{2}(\sigma_x \sigma_x + \sigma_y \sigma_y) = \sigma^+ \sigma^- + \sigma^- \sigma^+$

$H_I = J(\sigma^+ \sigma^- + \sigma^- \sigma^+)$: Another possible interaction type (flip-flop interaction)

The quantum compiler:

(Quantum Gates through spin rotations)

Remember that
 $U = e^{i\gamma} R_v(\theta)$

NOT gate:

$$X = R_x(\pi) = e^{-i\frac{\pi}{2}\sigma_x} = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}$$

$$X = -i \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

HAD. gate:

$$H = e^{-i\frac{\pi}{2}(\sigma_x + \sigma_z)/\sqrt{2}}$$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} -i & -i \\ -i & i \end{bmatrix}$$

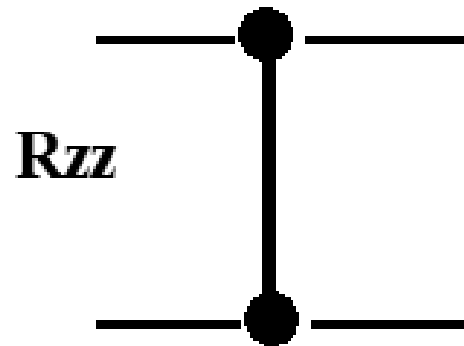
$$H = \frac{-i}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Z Gate:

$$Z = R_z(\pi) = e^{-i\frac{\pi}{2}\sigma_z} = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}$$

$$Z = -i \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Rzz Gate:

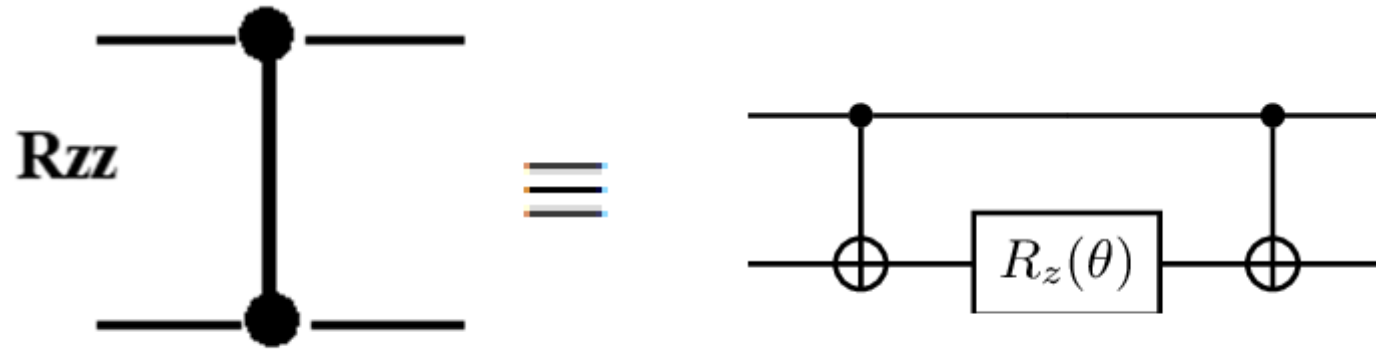


$$R_{ZZ}(\theta) = \exp\left(-i\frac{\theta}{2}Z\otimes Z\right) = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 & 0 & 0 \\ 0 & e^{i\frac{\theta}{2}} & 0 & 0 \\ 0 & 0 & e^{i\frac{\theta}{2}} & 0 \\ 0 & 0 & 0 & e^{-i\frac{\theta}{2}} \end{pmatrix}$$

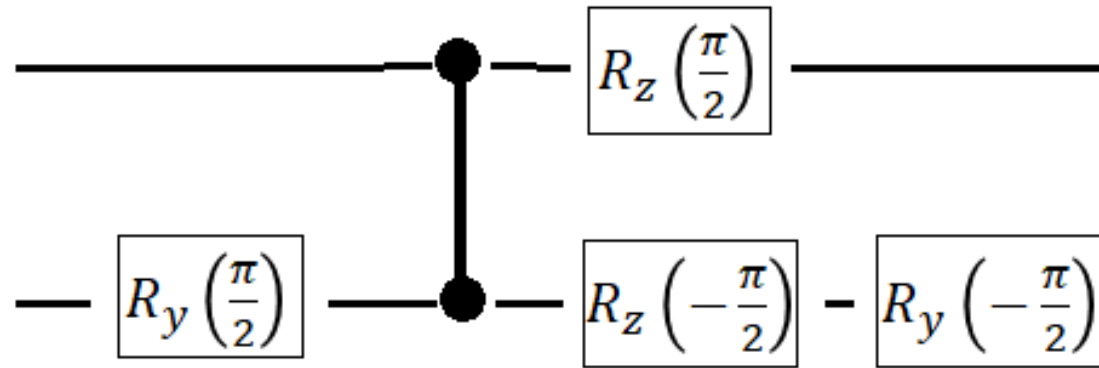
$$R_{ZZ}\left(\theta = \frac{\pi}{2}\right) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1-i & 0 & 0 & 0 \\ 0 & 1+i & 0 & 0 \\ 0 & 0 & 1+i & 0 \\ 0 & 0 & 0 & 1-i \end{pmatrix}$$

Decomposition:

Show that

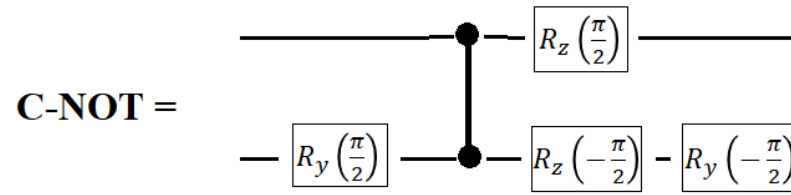


C-NOT =



$$R_z\left(\frac{\pi}{2}\right) = e^{-i\frac{\pi}{4}\sigma_z} \equiv Z$$

$$R_y\left(-\frac{\pi}{2}\right) = e^{i\frac{\pi}{4}\sigma_y} \equiv \bar{Y}$$



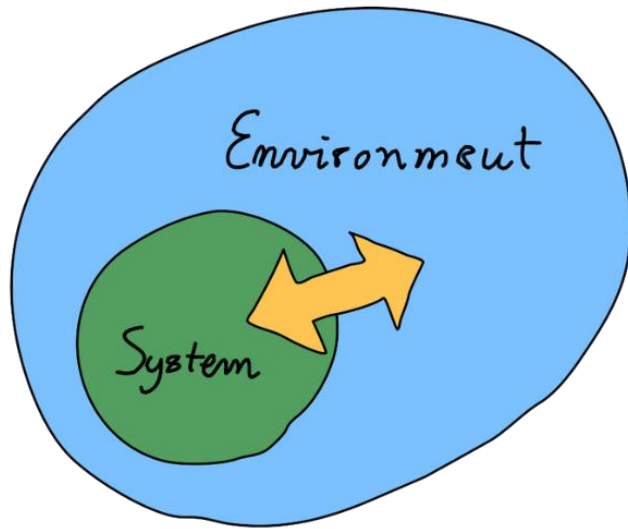
$$CNOT = (1 \otimes \bar{Y})(Z \otimes \bar{Z})(R_{ZZ})(1 \otimes Y)$$

$$(CNOT) = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1-i}{\sqrt{2}} & 0 & 0 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & \frac{1+i}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & \frac{1-i}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1-i}{\sqrt{2}} & 0 & 0 & 0 \\ 0 & \frac{1-i}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 & \frac{1-i}{\sqrt{2}} \\ 0 & 0 & \frac{1-i}{\sqrt{2}} & 0 \end{pmatrix} = \frac{1-i}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Open quantum systems And quantum noise:

So far, we have investigated the quantum dynamics for closed quantum systems.
But, actually no system is a closed system.



Open quantum system dynamics
are no longer unitary.

$$\rho(t) = \begin{bmatrix} p_e & c \\ c^* & p_g \end{bmatrix} \quad , p_e + p_g = 1$$

$$\rho(t) \xrightarrow{\text{Zero temp bath}} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\rho(t) \xrightarrow{\text{Finite temp bath}} \begin{bmatrix} p_e' & 0 \\ 0 & p_g' \end{bmatrix}$$

T_1 : Energy dissipation time

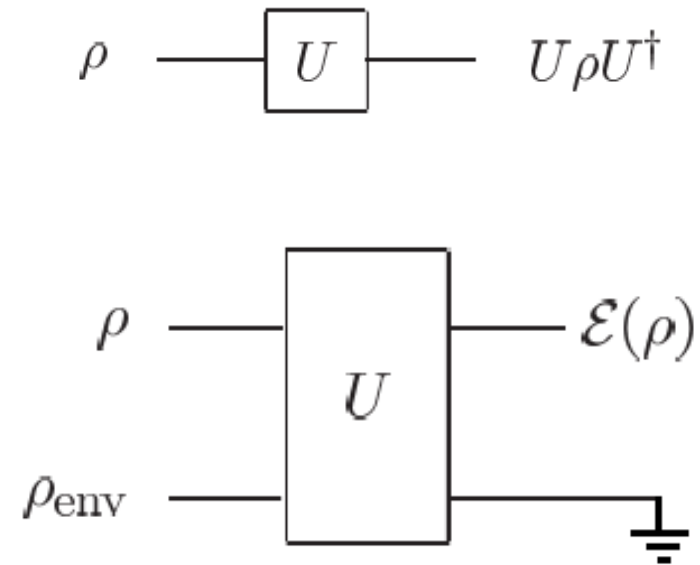
T_2 : Decoherence time

According to an approach, though the open quantum system evolution is not unitary, system+env. Evolution is unitary.

Then one can obtain the system of interest by tracing out the environment degrees of freedom (by partial trace).

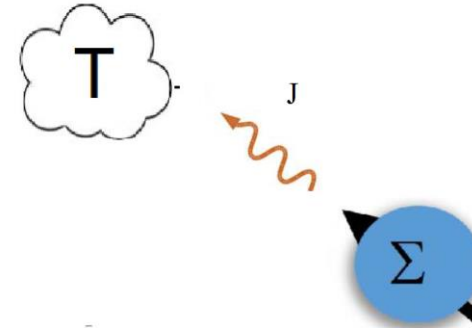
$$\mathcal{E}(\rho) = \text{tr}_{\text{env}} \left[U (\rho \otimes \rho_{\text{env}}) U^\dagger \right] .$$

The evolution described above is called a ‘Quantum operation’. This is a trace preserving dynamical map.



If the bath is a thermal bath, (finite temperature bath)
Then the bath (or the environment) can be described by a density matrix as

$$\hat{\rho}_{\text{env}} = \frac{e^{-\beta \hat{H}}}{Z}, \quad Z \equiv \text{Tr} \left[e^{-\beta \hat{H}} \right]$$



The system-bath Hamiltonian can be written as;

$$H_{SE} = H_{Sys} + H_{Env} + H_I$$

$$H_I = J(\sigma_S^+ \sigma_E^- + \sigma_S^- \sigma_E^+)$$

$$U = e^{-iH_{SE}t}, t : \text{Interaction time}$$

Simulating open quantum Dynamics:

$$\mathcal{E}(\rho) = \text{tr}_{\text{env}} \left[U (\rho \otimes \rho_{\text{env}}) U^\dagger \right] .$$

